

like the baby. However, such true AI applications will be achieved only in the far future (if at all).

Figure 1 shows that machine learning is a strict subset of AI and as such one of the many techniques used to implement artificially intelligent systems. Machine learning involves techniques in which large data sets are used to train programs so that the necessary task can be carried out effectively. Further, the accuracy of performing a particular task increases with larger and larger training data sets. Notice that there are other techniques used to develop artificially intelligent systems like Boolean logic based systems, fuzzy logic based systems, genetic programming based systems, etc. However, nowadays machine learning is the most vibrant technology used to implement AI based systems. Figure 1 also shows that deep learning is a strict subset of machine learning, making it just one of the many machine learning techniques. However, here again I need to inform you that, currently, in practice, most of the serious machine learning techniques involve deep learning. At this point, I refrain even from trying to define deep learning. Just keep in mind that deep learning involves the use of large artificial neural networks.

Now, what is data science (the red circle) doing in Figure 1? Well, data science is a discipline of computer science/mathematics which deals with the processing and interpretation of large sets of data. By large, how large do I mean? Some of the corporate giants like Facebook claimed that their servers could handle a few petabytes of data as far back in 2010. So, when we say huge data, we mean terabytes and petabytes and not gigabytes of data. A lot of data science applications involve the use of AI, machine learning and deep learning techniques. Hence, it is a bit difficult to ignore data science when we discuss AI. However, data science involves a lot of conventional programming and database management techniques like the use of Apache Hadoop for Big Data analysis.

Henceforth, I will be using the abbreviations AI, ML, DL and DS as shown in Figure 1. The discussions in this series will mainly focus on AI and ML with frequent additional references to data science.

The beginning of our journey and a few difficult choices to make

Now that we know the topics that will be covered in this series of articles, let us discuss the prerequisites for joining this tutorial. I plan to cover the content in such a way that any person who can operate a Linux machine (a person who can operate an MS Windows or a macOS machine is also fine, but some of the installation steps might require additional help) along with basic knowledge of mathematics and computer programming will definitely appreciate the power of AI, once he or she has meticulously gone through this series.

It is possible to learn AI in a language-agnostic way without worrying much about programming languages. However, our discussion will involve a lot of programming and will be executed based on a single programming language. So, before we fix our (programming) language of communication, let us review the top programming languages used for AI, ML, DL and DS applications. One of the earliest languages used for developing AI based applications was Lisp, a functional programming language. Prolog, a logic programming language, was used in the 1970s for the same purpose. We will discuss more about Lisp and Prolog in the coming articles when we focus on the history of AI.

Nowadays, programming languages like Java, C, C++, Scala, Haskell, MATLAB, R, Julia, etc, are also used for developing AI based applications. However, the huge popularity and widespread use of Python in developing AI based applications almost made the choice unanimous. Hence, from this point onwards we will proceed with our discussion on AI based on Python.

However, let me caution you. From here onwards, we make a number of choices (or rather I am making the choices for you). The choices mostly depend on ease of use, popularity, and (on a few occasions) on my own comfort and familiarity with a software/technique as well as the best of my intentions to make this tutorial highly effective. However, I encourage you to explore any other potential programming language, software or tool we may not have chosen. Sometimes such an alternative choice may be the best for you in the long run.

Now we need to make another immediate choice — whether to use Python 2 or Python 3? Considering the youth and the long career ahead of many of the potential readers of this series, I will stick with Python 3. First, let us install Python 3 in our systems. Execute the command ‘sudo apt install python3’ in the Linux terminal to install the latest version of Python 3 in your Ubuntu system (Python 3 is probably already installed in your system). The installation of Python 3 in other Linux distributions is also very easy. It can be easily installed in MS Windows and macOS operating systems too. The following command will show you the version of Python 3 installed in your system:

```
python3 --version
Python 3.8.10
```

We need to install a lot of Python packages as we proceed through the series. Hence, we need to install a package management system. Some of the choices include pip, Conda, Mamba, etc. I chose pip as our package management system for this tutorial because it is relatively simple as well as the recommended installation tool for Python. Personally, I am of the opinion that both Conda and Mamba are more powerful than pip and you are welcome to try them out. However, I will stick with pip. The command ‘sudo apt install python3-pip’ will install pip in an Ubuntu system. Notice that pip, Conda and Mamba are